PARISHRAM 2.0 2026 (Class 12th)

BIOLOGY (Botany)

Sexual Reproduction in Flowering Plants

DPP: 2

Q1 Which of the following statements correctly differentiate between syncarpous and apocarpous gynoecium?

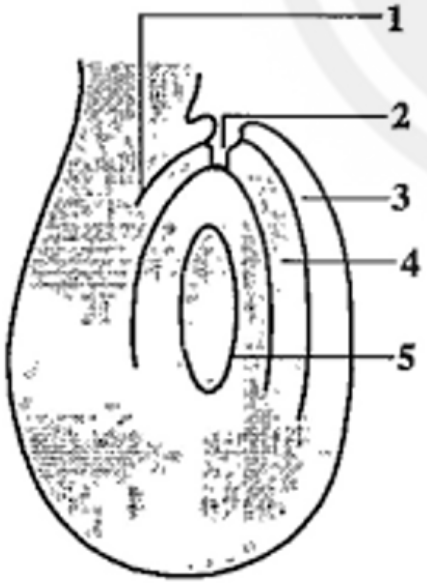
(A) A monocarpellary gynoecium always has multiple fused pistils, making it syncarpous.

(B) In a multicarpellary gynoecium, if the pistils are fused, it is called syncarpous, whereas if they are free, it is called apocarpous.

(C) An apocarpous gynoecium has a single pistil, while a syncarpous gynoecium has multiple pistils that are free.

(D) A syncarpous gynoecium consists of multiple pistils that are free, while an apocarpous gynoecium consists of multiple fused pistils.

Q2 Select the correct option for the given diagram.



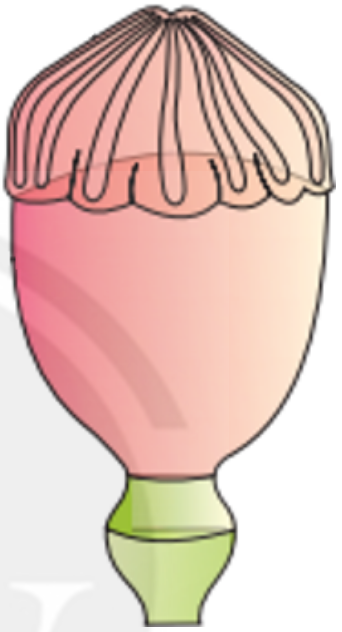
(A) 1- Hilum 2- Chalaza

(B) 3- Outer integument , 4- Inner integument

(C) 2- Chalaza 5- Funicle

(D) 1- Funicle 2- Micropyle

Q3 Select the correct option for the given diagram.



(A) Multicarpellary, syncarpous pistil of Papaver

(B) Monocarpellary , syncarpous pistil of Papaver

(C) Multicarpellary , apocarpous pistil of Papaver

(D) Monocarpelary , apocarpous pistil of Papaver

Q4 Select the correct option for the given diagram.



(A) Multicarpellary, apocarpous gynoecium of Marchantia
(B) Multicarpellary, apocarpous gynoecium of Michelia
(C) Multicarpellary, syncarpous gynoecium of Marchantia
(D) Multicarpellary, syncarpous gynoecium of Michelia

Q5 How many of the following are examples of many ovules ?
wheat, paddy, mango , papaya, water melon, orchids

(A) 4 (B) 3
(C) 5 (D) 2

Q6 Which of the following statements is incorrect regarding the structure of an ovule?
(A) The funicle connects the ovule to the placenta.
(B) The hilum is the junction between the funicle and the ovule.
(C) The micropyle is a small opening located at the chalazal end.
(D) The integuments surround the nucellus except at the micropyle.

Q7 Which of the following statements about the structure of an ovule is incorrect?
(A) The nucellus is enclosed within the integuments.
(B) The nucellus contains abundant reserve food materials.
(C) The embryo sac is located outside the nucellus.
(D) The embryo sac develops from a megaspore.

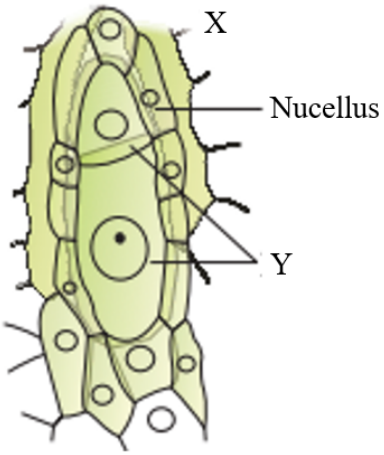
Q8 Match the **List-I** and **List-II**

List-I	**List-II**
P Monocarpellary	1 Pistils are fused
Q Multicarpellary	2 More than one pistil
R Syncarpous	3 Pistils are free
S Apocarpous	4 Single pistil

(A) P-4 Q-2 R-1 S-3
(B) P-3 Q-2 R- 1 S-4
(C) P-2 Q-1 R-4 S-3
(D) P-1 Q-2 R-3 S-4

Q9 **Assertion:** The megaspore mother cell (MMC) leads to the formation of four megaspores.
Reason: MMC undergoes meiosis during megasporogenesis.
(A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.
(B) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(C) Assertion is true, but Reason is false.
(D) Assertion is false, but Reason is true.

Q10 Choose the correct labelling for the given diagram.



(A) X- Chalazal end Y- Megaspore tetrad
(B) X- Chalazal end Y- Megaspore dyad
(C) X- Micropylar end Y- Megaspore dyad
(D) X- Micropylar end Y- Megaspore tetrad

Answer Key

Q1	B
Q2	B
Q3	A
Q4	B
Q5	B
Q6	C
Q7	C
Q8	A
Q9	A
Q10	C

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

In a multicarpellary gynoecium, if the pistils are fused, it is called syncarpous, whereas if they are free, it is called apocarpous.

Video Solution:

Q2 Text Solution:

3- Outer integument , 4- Inner integument

Video Solution:

Q3 Text Solution:

Multicarpellary, syncarpous pistil of *Papaver*

Video Solution:

Q4 Text Solution:

Multicarpellary, apocarpous gynoecium of *Michelia*

Video Solution:

Q5 Text Solution:

papaya, water melon, orchids

Video Solution:

Q6 Text Solution:

The micropyle is a small opening located at the chalazal end.

Video Solution:

Q7 Text Solution:

The embryo sac is located outside the nucellus.

Video Solution:

Q8 Text Solution:

P-4 Q-2 R-1 S-3

Video Solution:

Q9 Text Solution:

Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

Video Solution:

Q10 **Text Solution:**

X- Micropylar end Y- Megaspore dyad

Video Solution:

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